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Machine Learning Techniques for Predicting Ovarian Cancer in its Early Stages Using Biomarkers

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***Abstract*—The deadliest gynecological cancer affecting women is ovarian cancer, currently incurable with no effective medication treatments. The key focus of this research is to assess insights for early diagnosis using statistical analysis and machine learning techniques on data from clinical trials obtained from 349 patients. Several techniques, including Random Forest, Decision Tree, Gaussian NB, AdaBoost, and Logistic regression, were applied to find the most reliable factor for ovarian cancer prediction. A clinically evaluated raw dataset of benign samples and malignant ovarian tumor patient data set is used to develop early-stage ovarian cancer predictions, and the effectiveness of ML models was examined utilizing metrics including F1-score, Accuracy, Precision, and Recall. The proposed study shows better outcomes, with the Random Forest classifier exhibiting the highest accuracy for validation at 99% based on the test data of ovarian cancer predictions. Even though early-stage ovarian cancer detection is generally unavailable, cancer diagnosis may be greatly aided by machine learning.**

Keywords— Ovarian-Cancer, Machine Learning, Feature Selection, Random Forest

I. INTRODUCTION

Ovarian cancer(OC)is among the commonly encountered cancer that is the sixth leading cause of death among females, with lifetime risk factor at 2.7% which is estimated to 207,252 fatalities and 313,959 cases globally in 2020 [1, 2].The reproductive glands in women are called ovaries which contain three different kinds of cells: stromal, germ, and epithelial. Ovarian tumors can originate from aberrant cell growth and be either non-cancerous or malignant. Cancer cells can migrate from the ovaries across the pelvic region and continue to expand to the abdominal area and other organs if malignant tumors fail to be detected at an early stage .Identifying the stage of ovarian cancer is essential for designing a treatment plan.

Depending on the type of cell in which it begins, about thirty distinct forms of ovarian cancer exist. Approximately 85–90% of these tumors begin in the cells that coat the outer surface of the ovary. There are several forms of cancer of the ovary. The three primary cell types that comprise the ovary are named after from which they originate [3]. They are: Surface epithelial cell "Epithelial Tumors", Germ cell "Germ Cell Tumors", and Ovarian stroma "Sex-cord Stromal Tumors". The specific reason why ovarian cancer develops is still unknown but there are many factors believed to be involved such as: Age, reproductive history, obesity, drugs, Assisted reproduction, history of ovarian cancer in the family, cancer syndromes, talcum powder, smoking, etc.

The majority of ovarian cancer patients are asymptomatic when their symptoms initially appear, even If they cause signs, the signs are imprecise and complex to detect. Consequently, ovarian cancer becomes the "silent killer" since patients are typically diagnosed at an advanced stage [4]. Usually, ovarian cancer symptoms appear once the disease has advanced past the ovaries, however, in some people, symptoms may appear at any stage of the disease.

Ovarian cancer symptoms are sometimes mistaken since they might be mild encountered by women who are not impacted by the illness. They are often mistaken with symptoms of less serious disorders, particularly digestive problems, resulting in frequent misdiagnosis. Most people are only discovered in the latter stages of the disease, when treatment becomes increasingly difficult.

Approximately 90% of women, regardless of the type or stage of ovarian cancer, had more than one symptom before being diagnosed. Prior to their diagnosis, more than two-thirds of women had never heard of ovarian cancer or knew anything about it. Improving women's understanding of ovarian cancer and its symptoms is critical if we are to reduce diagnostic delays.

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This is particularly noteworthy because there is currently no well recognized, straightforward diagnostic procedure for ovarian cancer detection[5].

Although several studies have performed to detect ovarian cancer, evaluation regarding accuracy aresufficient,yet there is still room for development. Furthermore, the primary goals of our methodology are as follows:

- The proposed study established an early-stage ovarian cancer prediction process using biomarkers.
- Six machine learning techniques were employed to identify significant and related biomarkers.
- ANOVA feature selection technique and standard scaling are used to find noteworthy characteristics and prevent redundant data.
- The Random Forest classifier is the best fit for predicting ovarian cancer and achieved 99% accuracy.

II. Related Work

Lu.M. et al. [6] used machine learning to forecast ovarian cancer (OC) and benign ovarian tumors (BOT) in Chinese patients. In both training and testing data, a decision tree model delivers higher prediction results. The findings are also equivalent to the logistic regression model, which employs both the HE4 and CEA characteristics.

Akazawa, M et al. [7] created a model based on artificial intelligence (AI) to forecast the pathology diagnosis of ovarian cancers. Five machine learning classifiers were utilized in this study and the highest accuracy attained by the XGBoost approach was 0.80.

Garg, R et al [8] used machine learning and natural language processing (NLP) on preoperative CT scans to enhance the capacity to forecast surgical problems and hospital readmissions in women receiving debulking surgery for ovarian cancer. Ahmed, M.R.et al. [9] examined ovarian cancer data from 521 women in order to determine significant risk factors. The data was collected from Hospitals of Dhaka and analyzed using various machine learning algorithms, with Logistics Regression providing the maximum accuracy of 0.933.

Kawakami, E. et al [10] discussed the application of machine learning approaches for forecasting many clinical characteristics related to epithelial ovarian cancer (EOC). Yuan, Y. et al. [11] created a machine learning model utilizing the XGBoost algorithm to forecast lung metastatic risk in individuals with ovarian cancer who have just received a diagnosis. With an accuracy rate of 0.982, sensitivity of 1.000, specificity of 0.686, and AUC of 0.843, the model demonstrated strong discriminative abilities.

C. Cao et al. [12] analyzed data from the SEER database patients diagnosed with ovarian cancer between 2010 and 2016.The variables linked to the development of LM were identified using univariate and multivariate logistic regression analyses, and the overall survival of OC patients with LM was estimated through Kaplan-Meier analysis. J. Xu et al. [13] investigated the occurrence of lung metastases (LMs) and survival outcomes in LM-positive individuals with epithelial ovarian cancer (EOC). The study analyzed data from 33,418 qualified EOC patients and found that 2240 (6.7%) LMs while diagnosed EOC.

T. Zhang et al. [14] analyzed data from 291 women and found that within 30 days, 25 patients (8.6%) required readmission, while 45 patients (15.5%) experienced postoperative problems. The patients were 59 years old on average, with a preoperative mean CA125 value of 610 U/ml and an albumin level of 3.9 g/dl.

Liu, S.Y. et al [15] explored the landscape of the tumor microenvironment (TME) and its predictive value in epithelial ovarian cancer (EOC) using somatic mutation analysis, immune infiltration analysis, and survival analysis.

Shannon NB, et al. [16] study created a signature that predicted cisplatin sensitivity using information from The Cancer Genome Atlas and Genomics of Drug Sensitivity in Cancer. Four putative biomarkers (CYTH3, GALNT3, S100A14, and ERI1) were discovered, refined for immune histochemistry (IHC) application, and confirmed in a group of patients undergoing adjuvant carboplatin and surgical resection.

While previous studies by Akazawa et al. [7] and Kawakami et al. [10] leverage machine learning models for predictive modeling in ovarian cancer, Akazawa et al. identified a gap in the absence of comprehensive data preprocessing and feature selection strategies, despite achieving an accuracy of 80%.This gap aligns with our aim to enhance the robustness and accuracy of predictive models Additionally, while Kawakami et al. utilize K-fold cross-validation, the incorporation of additional evaluation metrics such as F1 Score and AUC-ROC curve could provide a more comprehensive understanding of classifier performance. This gap underscores the importance of improving model assessment techniques in our research. Furthermore, the studies by Yuan et al. [11] and Zhang et al. [14] lack evaluation metrics and calibration, suggesting the need for improved model assessment. Exploration of feature selection techniques and dimensionality reduction methods could further enhance the predictive models developed in these studies.

III. Methodology

The raw dataset consists of two categories of data: patients with benign ovarian tumors and patients diagnosed with ovarian cancer. In order to generate better results, both data sets must be processed. Utilizing preprocessing is followed by feature optimization following preprocessing, both data sets should undergo feature optimization using the suggested approach. Following feature optimization, various ML model are used to ascertain the capable of yield the most optimal classifications. The recommended approach ultimately employed the most effective ML methods. **Fig. 1** depicts the workflow of early detection of ovarian cancer. This diagram depicted all of the procedures, from data preparation to performance analysis.

A. Dataset Collection

Systematically gathered from the third Affiliated Hospital of Soochow University, the dataset includes information from 349 patients [17].The examination of the past spanned from July 2011 to July 2018, encompassing a cohort of 171 patients diagnosed with OC 178 patients with benign ovary tumors. The

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pathology diagnosis was used to gather the dataset's 49 attributes.

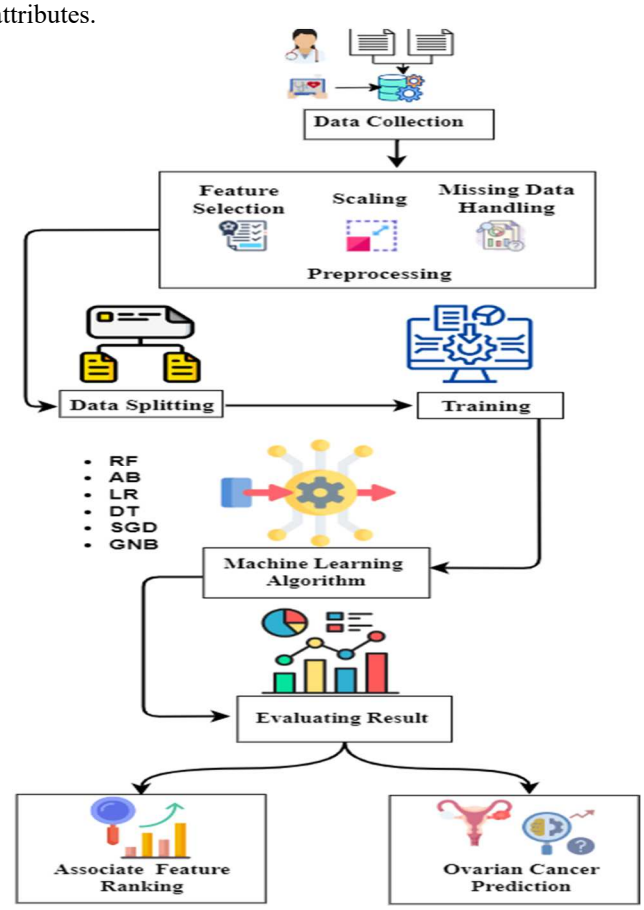


Fig.1: Proposed model of our ovarian prediction system.

B. *Correlation Coefficient*

The notion of correlation functions as a measurable measure that characterizes the relationship or dependency between two variables.. That is, if one variable's value increases or decreases, the measurement of how much the value of the other variable will increase or decrease is expressed through correlation. Correlation values are scaled, ranging from -1 to 1. The link between the two variables is stronger when it is closer to 1, and weaker when it is closer to zero. The relationship between the two factors K and M is [18] described as:

$$\rho_{K,M} = \frac{cov(K,M)}{\sigma K \sigma M} \qquad (1)$$

C. *Data Preprocessing*

Preprocessing is a process that converts the unprocessed data into an understandable and relevant shape. Numerous machine-learning techniques were applied to do data cleaning and preprocessing [19]. To deal with inconsistent and incomplete data, a variety of preprocessing methods are used, such as handling missing values, identifying outliers, discretizing the data and reducing both dimensionality and numerosity. In this approach, the missing column is replaced using the mean approach and converted into categorical data into numerical data using a label encoder. Then used feature Selection for removing irrelevant data.

D. *Label Encoding*

One common encoding technique that works with categorical values is label encoding. According to this method, every label obtains a unique number accordance with order of letter. The following formulae are used to calculate the encoded category values:

$$S = \frac{1}{1 + \exp(-\frac{n - mdl}{a})} \qquad (2)$$

$$\hat{x}^k = prior * (1 - s) + s * \frac{n^+}{n} \qquad (3)$$

Here, the power of regularization is represented by a=Smoothing parameter, where mdl=min data (samples) in leaf. MDL and a are recommended to have values between 1 and 100. The previous values are substituted by new ones that only exist once in the train dataset [20].

E. *Data analysis*

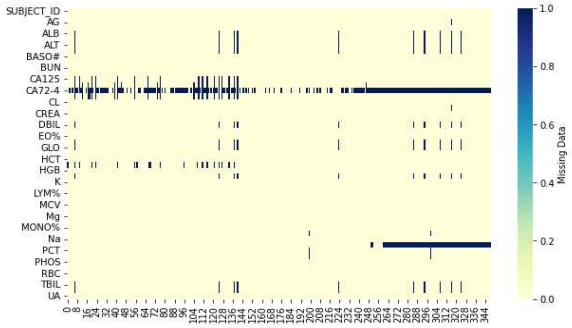
Verifying the dataset is accomplished through data analysis. Data must be cleaned, transformed, and modeled to find patterns, trends, and connections. The objective of data analysis is to uncover relevant information that can aid in decision-making. For instance, the analysis might assist in examining the qualities of the data and the correlation among features.

F. *Missing Data Handling*

The Clinical dataset (349 patients with 49 features)includes several missing data. In an attempt to impute missing data, this study used the mean method. This allowed researchers to determine the average of and substitute the mean values for the missing values.

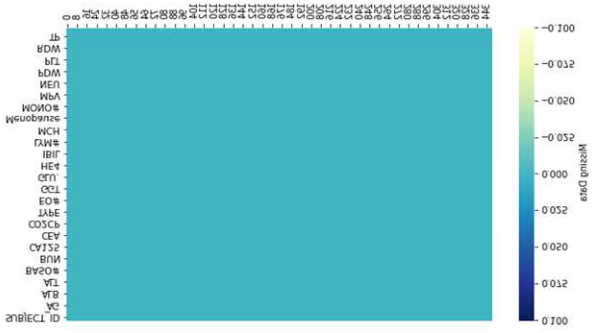
$$\text{Mean} = \frac{\text{Sum of all data points}}{\text{Number of data points}} \qquad (4)$$

In the column “AFP,” there are 22 rows with missing values, and "CA19-9” have twenty-four missing rows. Process of substituting equal values for missing values is known as imputation. The mean value is used to replace the missing values. **Fig. 2**.depicts the data visualization before and after missing data processing. “CA125” have seventeen row missing data, and “CA19-9” have twenty-four rows. Missing values imputed using mean approach.



(a) With Missing Data.

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(b) Without Missing Data

Fig.2: Data visualization of used dataset (a) With Missing Data. (b)Without Missing Data

G. *Feature Selection*

Feature selection is selecting related attributes by shrinking the scale of input property. Objective of the strategy is to reduce the amount of input parameters, which helps a model forecast and raise the accuracy score by minimizing the dataset's identity. This study uses feature selection in ovarian cancer clinical data. The study employed the select k best strategy to determine the top feature, and for scoring function ANOVA F.

H. Standard Scaler

A StandardScaler is a method used to standardize the features of a dataset by transforming them so that their standard deviation is one and their mean is zero. The mathematical representation of Standard Scaler is given by the following equation.

$$X_{new} = \frac{X_i - X_{mean}}{X_{sd}} \tag{5}$$

Where X_{new} is the original feature, X_i is the standardized feature,$X_{mean}$is the mean of the feature, and X_{sd} is the standard deviation of the feature.

I. *Data Splitting*

The most common way to split a dataset is into three subsets: training, validation, and testing. In addition to using tenfold cross validation, the study employed 80% of the data for training and 20% for testing. The model is trained on the training set, and its performance during training is assessed and its hyperparameters adjusted on the validation set. The model's final performance was assessed using the test dataset.

J. *Random Forest*

Random Forest is an ensemble learning technique. This kind of supervised machine learning technique is applied to regression and classification problems. Combining the predictions from several decision trees minimizes variance and raises the model's overall accuracy [21]. To determine the classification's Gini Index, use the following formula

$$Gini = 1 - \sum_{J-1}^{n} (p_J)^2 \tag{6}$$

The value of p_j in equation (6) represents the likelihood that an object will be assigned to a particular class or feature. The model's hyperparameters were set with a random_state of 42 during training, ensuring reproducibility, while the n_estimators parameter was fixed at 100, representing the number of trees in the Random Forest ensemble. These hyperparameters were determined to constitute the optimal configuration for the model based on the characteristics of the dataset.

IV. Result and Discussion

Several performance indicators, such as the F1 score, recall, accuracy, and precision, are used in this work. Using a 10-fold cross-validation method, the optimal parameters for each model are found, and the accuracy of each model is then evaluated. Overfitting and factor adjustment concerns have been observed using a range of metrics and methods. Based on Python-3, the Scikit-learn libraries, settings, and resources were used in this study to carry out its experiments and analyze the data.

A. *Confusion matrix*

The results are assessed in terms of recall, f-measure, accuracy, and precision using a confusion matrix and classification report. The result depends on how accurately the model was trained. True positive (TP), false positive (FP), true negative (TN), and false negative (FN) counts were used to calculate the performance assessment indicators shown in eq. (7) to eq. (10).

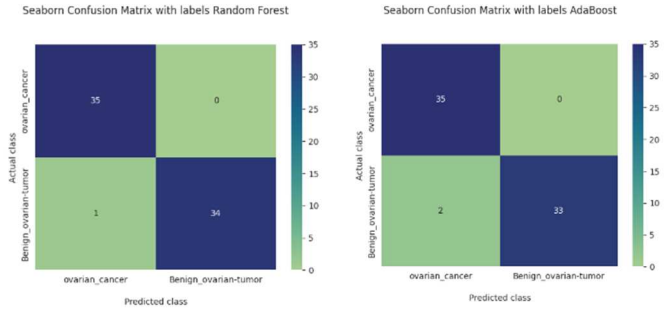
$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{7}$$

$$\text{Precision} = \frac{TP}{\text{TP} + \text{FP}} \tag{8}$$

$$\text{Recall} = \frac{TP}{\text{TP} + \text{FN}} \tag{9}$$

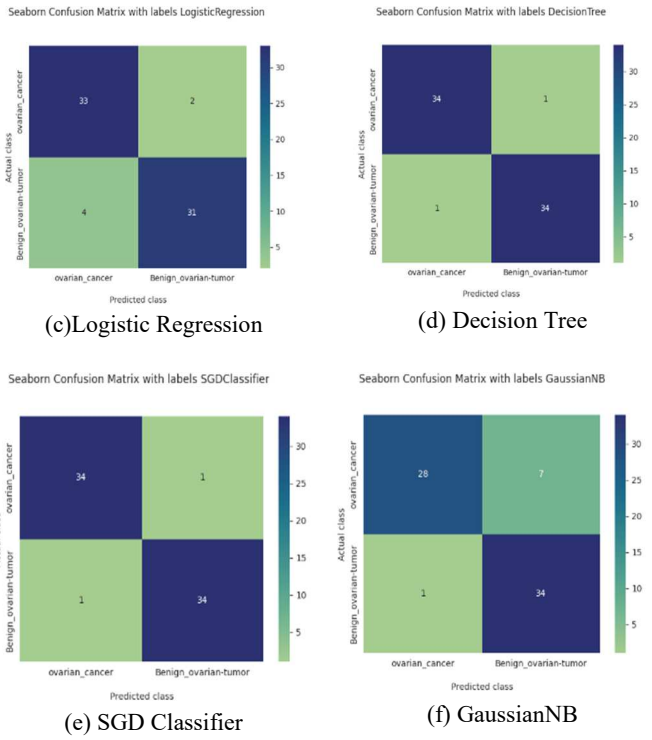
$$\text{F1} = 2\frac{Precision.Recall}{Precision + Recall} \tag{10}$$

Precision and recall are the two primary performance criteria utilized in medical diagnostics research. The F-measure, which combines recall and accuracy into a single, easier-to-use value for comparisons, is especially significant for reporting performance in medical diagnostics. Within the medical field, recall is considered a more important parameter since false negative results usually cause more harm than false positive results. A patient may suffer greatly from a missing condition, but since doctors can dismiss FPs (precision), missed FNs (recall) may not be as serious. In light of this, an F1 statistic that has been modified to prioritize recall over precision is more helpful. **Fig. 3**. displays the confusion matrix of machine learning model.



(a) Random Forest (b) AdaBoost

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(c)Logistic Regression (d) Decision Tree

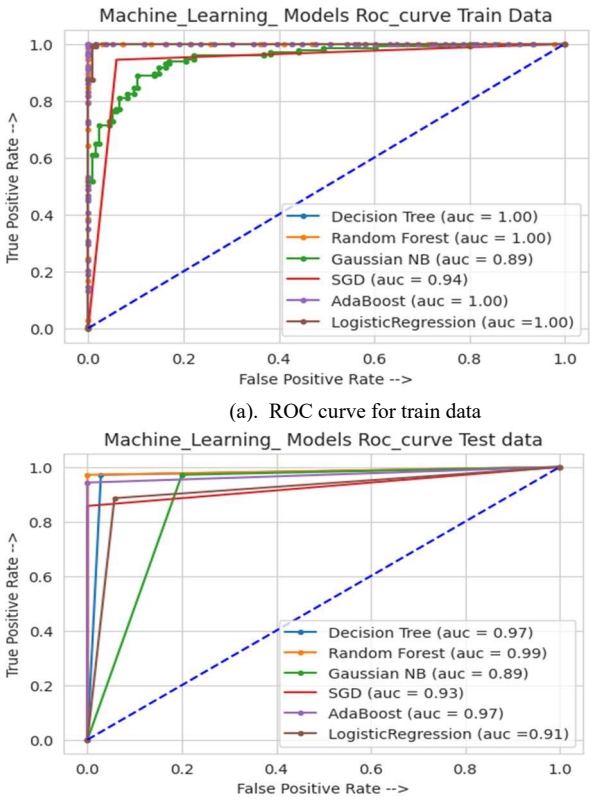
(e) SGD Classifier (f) GaussianNB

Fig.3: Confusion Matrix of (a) Random Forest (b) AdaBoost(c) Logistic Regression (d) Decision Tree (e) Stochastic Gradient descent (f) GaussianNB.

In order to reduce overfitting, **Table I** compares the accuracy of each model's testing and training data. Each model's precision, recall, accuracy, and F1-score are displayed.**Fig.4** (a) denotes ROC curve of different machine learning classifiers for testing data and **Fig. 4** (b) shows ROC curve for training data.

TABLE-I: PERFORMANCE EVALUATION OF SEVERAL ML MODELS

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	99%	0.99	0.99	0.99
AdaBoost	97%	0.97	0.97	0.97
Logistic Regression	91%	0.92	0.91	0.91
Decision Tree	97%	0.97	0.97	0.97
Stochastic GradientDescent	97%	0.97	0.97	0.97
GaussianNB	89%	0.90	0.89	0.88



(a). ROC curve for train data

(b). ROC curve for test data

Fig.4 ROC curves of different machine learning classifiers.

Fig.5 shows the results of the test dataset on ovarian cancer, where the model opted for the attributes with the lowest squared error. Through SGD learning, all of the missing values were eliminated and the nominal attributes were converted to binary ones. Additionally, MLP learning was normalized for each attribute.

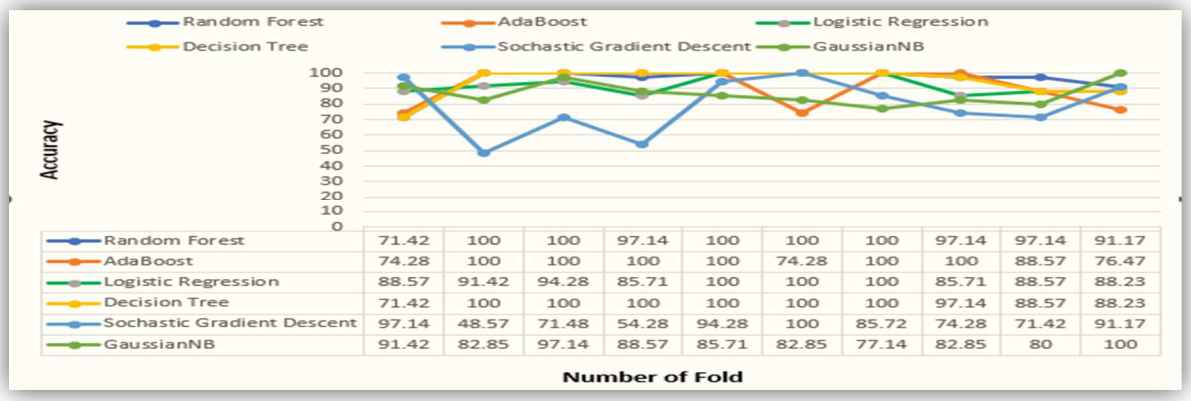


Fig.5 Accuracy comparison of different classifiers

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Table II compares our research's findings with previous ML based OC prediction work, demonstrating the superiority of our research. By applying DT, Lu et al. [6] were able to acquire an accuracy of 87.00% and a SE of 82.00%, respectively, on clinical data comprising 49 characteristics and 349 patients. In a different study, Akazawa et al. [7] achieved an accuracy of 80.00% by using XGBoost. Furthermore, Martinez-Mas [22] used ELM and SVM classifiers and achieved 87.00% accuracy, 87.00% SE, and 89.00% AUC on image data. In contrast, we used RF approaches in our proposed study and attained the best accuracy of 99%, which is superior than the prior results. It concludes that, in comparison to all other approaches, our proposed approach performs better.

Table-II: COMPARISON TABLE WITH EXISTING WORK

Author	Dataset	Method	Accuracy
Lu et al. [6]	Clinical data (349 patients With 49 features)	DT	87%
Akazaw et al. [7]	Clinical data (202 patients with 32 features)	XGB	80%
Ahmed et al. [9]	Customs Data(47 factor)	LR	93%
Kawakami et al. [10]	435 Patients (334 patients with EOC and 101 patients with benign ovarian tumor)	RF	96%
Yuan et al [11]	16059 patients with ovarian cancer(SEER) database	XGB	98%
Shannon et al. [16]	Genomics of Drug Sensitivity in Cancer (GDSC) and The Cancer Genome Atlas (TCGA)	XGB	82%
Proposed method	**Clinical data (349 patients with 49 features)**	**RF**	**99%**

V. CONCLUSION

In the context of our proposed investigation, we achieved an outstanding 99% peak accuracy using Random Forest (RF) methods, outperforming previous studies. The conclusion drawn is that, when compared to alternative approaches, our proposed method demonstrates superior performance. The primary contribution of this work lies in its utilization of select k best for feature selection, addressing feature significance, scaling, and data imputation for missing data. Validation is carried out through k-fold cross-validation in this study. While some studies have identified optimal models, this article highlights the Random Forest model's superiority in predicting maximum accuracy (99%). Consequently, the study strongly recommends the development of a Random Forest-based method for predicting ovarian cancer.

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